

- 8 Snow leopards (*Panthera uncia*) live in mountainous regions in Asia.

The photograph shows a snow leopard on a snow-covered mountain.



(Source: © Natural Visions/Alamy Stock Photo)

- (a) Give **one** reason for a named anatomical adaptation of this snow leopard that enables it to occupy its niche.

Use the information in the photograph to support your answer.

(1)

Adaptation

Reason

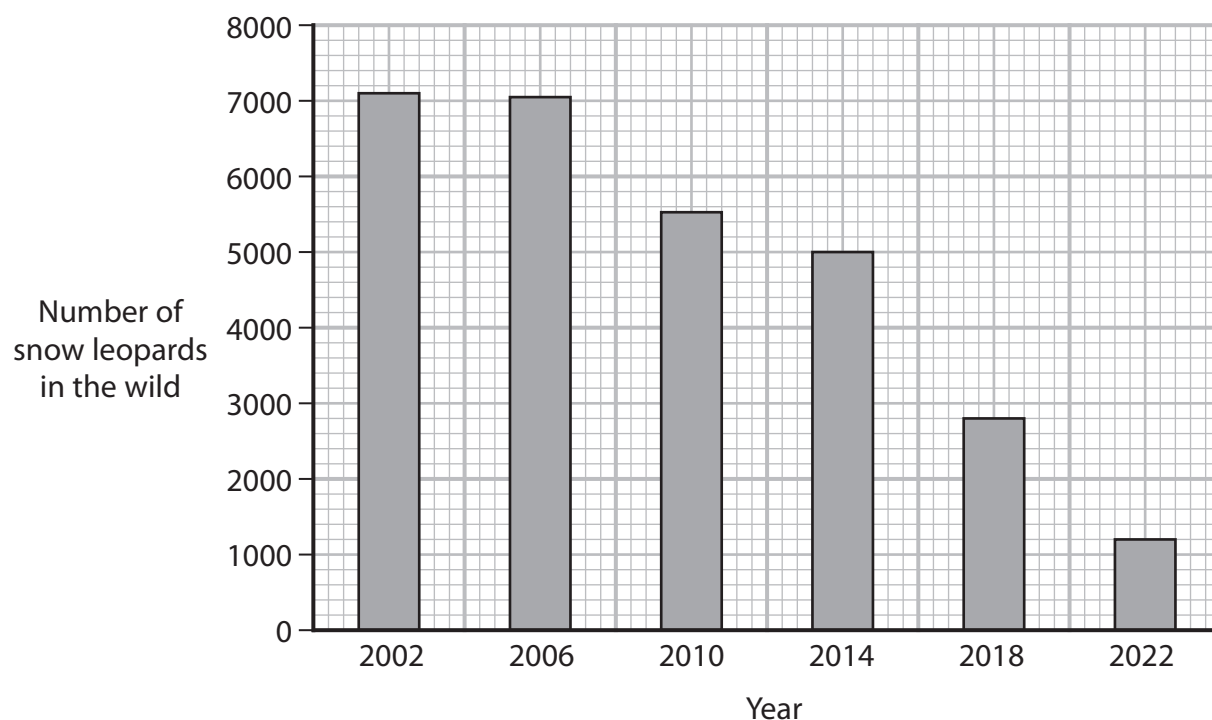
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(c) The graph shows the numbers of snow leopards in the wild since 2002.



(i) Calculate the percentage decrease in snow leopards from 2002 to 2022.

Give your answer to one decimal place.

(2)

.....%

(ii) Suggest **one** reason for this decrease in the number of snow leopards.

(1)

.....

.....

.....



- (d) Populations of snow leopards, living in different areas of Asia, sometimes develop different characteristics. These populations are called subspecies.

Individuals from different subspecies are able to produce fertile offspring.

Some scientists have proposed that there are three subspecies of snow leopard.

The table shows the names and locations of the snow leopard range for the three subspecies.

Subspecies	Location of snow leopard range
<i>P. uncia irbis</i>	Northern
<i>P. uncia uncia</i>	Western
<i>P. uncia uncioides</i>	Central

- (i) The northern and central locations are separated by a desert region.

The width of the region is shown by the line labelled X on the map.

Calculate the width of this desert region using the scale shown on the map.

Give your answer in standard form.

(2)

..... km

*(ii) Explain how these subspecies of snow leopard could form.

Use the information in the question to support your answer.

(6)

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- (iii) Give a reason as to why these different populations are described as subspecies and not different species of snow leopard.

(1)

(Total for Question 8 = 16 marks)

TOTAL FOR PAPER = 80 MARKS

